

508 paid FTE per month to meet service level

1

THE ANSWER — JULY TO SEPTEMBER

508 paid FTE per month

To hold **80% / 20s** phone and **80% / 60s** chat at the June run-rate +**10%** seasonal uplift.

PRODUCTIVE HOURS

68,827
per month — the BPO bill

CONTACTS

294k
all languages, all channels

AHT actual vs assumption **±78 FTE**

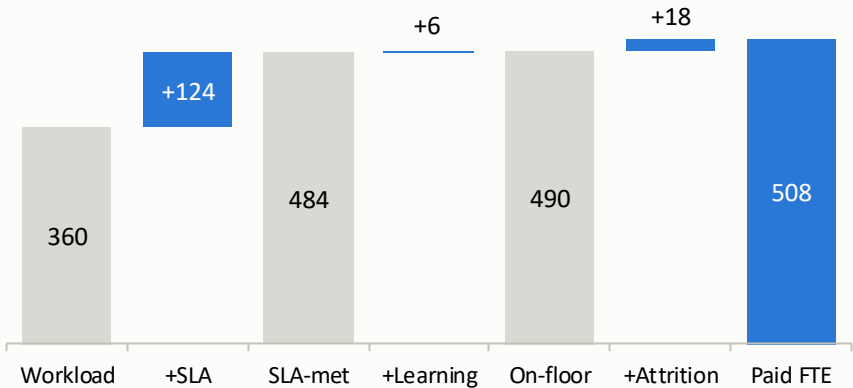
When channel AHT assumption is compared with English actual AHT then we see significant difference **710s** assumed vs **909s** actual. Depending upon which AHT is used there is swing of **78 FTE**

Assumption taken

- 1. Occupancy: 85%
- 2. Chat concurrency: 1.2 for all langs
- 3. July/Aug/Sep: 1.10 of June contact volume (no month-on-month variation)
- 4. Many data quality issues were spotted (in Appendix)

2

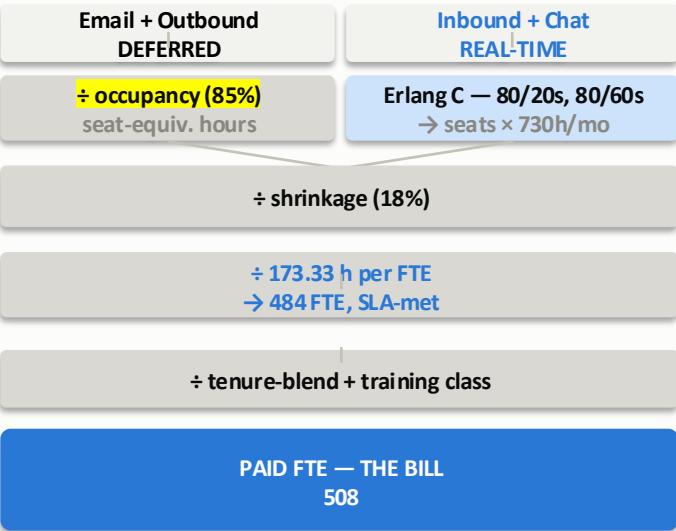
A QUARTER OF THE REQUIREMENT IS SERVICE LEVEL, NOT WORKLOAD



A workload model returns **360 FTE** and **misses SLA** — it implicitly assumes 100% occupancy. Erlang C says English inbound runs at **70%**; the idle 30% is the 20-second answer time.

3

MODEL ARCHITECTURE — ONE CHAIN, TWO PATHS IN



Two paths — deferred (workload) and real-time (Erlang C) — merge before shrinkage. Same chain, every language. Full formula and constants on slide 03.

Multiple strategies to optimize

Lever	FTE (per month)	Dependency	Action today, alone?	Details
BPO1/BPO2 channel routing (English)	4.5	None	Yes	Splitting 50/50 on every channel sends some Inbound/Outbound calls to the slower vendor (BPO2) and some Chats to the slower vendor (BPO1), for no reason — both are already contracted for English Action: route Inbound + Outbound fully to BPO1, Chat fully to BPO2 (Email either). Pure IVR/chat-router config change.
Pool Spanish + French (pilot)	18.8	Confirm bilingual agents exist at BPO3	No — one check first	Core mechanism: small non-English queues (DE/IT/ES/FR) are staffed 24/7 and each independently pays the 80/20 minimum-seat floor even when volume is low. Pooling shares that floor across languages instead of paying it once per language.
Pool all 4 non-English (phase 2)	51.4 cumulative	Pilot validates first	No	
Add English to the pool (phase 3)	78.8 cumulative	Vendor consolidation (BPO1/2 → BPO3)	No	
Shrinkage restagger (Break+Lunch off-peak)	5.8–11.5	Labor/scheduling buy-in; model has no intraday data to prove it	No	Shrinkage is 18% everywhere 8 of the 18 points are scheduling choices rather than entitlements, in theory movable off peak-demand hours.
Chat concurrency verification	up to 75	1.2 chat concurrency only for English, is it also same for other langs?	-	

How AI was used to solve this case study

Automation: Excel file was used as an input and all subsequent steps like data cleaning, configuration, forecast, capacity plan are generated by python files. If you include/exclude any data, this tool would give the plan accordingly.

Validation: Build a code through AI is good until is validated by a different AI model, therefore 2 AI models have been used

Quality: In my 1st draft there were couple of issues in AI output. E.g. **SLA is computed but never costed. The FTE number does not meet service level.** It suggested an optimization strategy to move **phone to chat** but in reality chat AHT is more than phone therefore AI suggestion was not correct. Through a QA process I mentioned to highlight these things

What was not done: I didn't just give AI the data and asked what is the forecast and capacity. I created a framework which I could test, reason and validate.

What could have been better: Given it is a synthetic data therefore in validation and doing a quality check it hallucinate a lot. E.g. outbound and inbound contact volume is equal which would generally be the case

THE MODEL	
MODULE	RESPONSIBILITY
config.py	Every assumption in one frozen dataclass
erlang.py	Erlang B/C, service level, ASA, solver
loader.py	Schema-driven reader; refuses the hidden sheet
clean.py	12 data-quality checks → structured log
forecast.py	Break detection , growth test, scenarios
capacity.py	Workload → Erlang → seats → FTE, by BPO
optimise.py	Each lever re-solved through the same engine
report.py	16-sheet Excel export

VERIFICATION — 72 CHECKS, ALL PASSING	
LAYER	WHAT IS ACTUALLY PROVED
Erlang B/C	Recursion vs closed form (1.1e-16); vs stationary dist. (<1e-12)
Simulation	vs discrete-event M/M/c — 3σ band on SL and ASA
Solver	Minimal feasible seats; SL monotone in seats
Queueing	Little's Law holds exactly
Raw cells	Full July chain re-derived from cells — matches 1e-6
Invariants	Every queue ≥80%; roll-up sums exact; LOW≤BASE≤HIGH
HR maths	Attrition compounds to 36.00%/yr
Levers	Each 1b strategy re-solved, not asserted

Leading the transformation — and how AI was actually used

2A · 30 / 60 / 90

0–30 · Understand the business (most important)

Before doing any development, I would spend first 30 days in understanding the business. A capacity plan may look great in theory, but many constraints would make that plan fail fast therefore

- Meet care experience and strategy **team** and **stakeholders**, understand where we stand right now, what are the goals for the future, what are the bottlenecks, what is good/can be improved, understand the constraints we work with, how AHT assumptions are calculated, how BPO agents lang skillset is distributed, How is current calls are distributed across channels, how off-peak and peak intraday hours are managed, leverage all existing on-boarding documentation if available, understand current forecast framework, how AI is currently used within the team, what is the team's current core skillsets
- **Outcome expected:** Build a business understanding document and align with Nils (head of care experience) to share the learning on what's good and what could be improved. This document also acts as a on-boarding document for other team members, so it is not just sitting in silo

31–60 · Act on improvement areas

Rather than fixing everything at once, I would break down the improvement areas based on couple of factors such as Data Complexity, Effort required, Dependency on other team/area, Impact

Identify high impact areas with low/medium effort or complexity which would be the quick wins. E.g. if we are using excel workbook as an input, I would try to find a solution based on my learning if it could be moved to a database or we could directly fetch this information from data

61–90 · Benchmark and compare

Main KPI I am looking to improve here is **reduce time to build the forecast with same/better quality**. If earlier we used to take 1 day to build a plan, and if within 3 months we can reduce that to 4 hours/ 3 hours with quick wins I believe that would not only motivate the team but overall bring better stakeholder experience. I would keep the realistic expectation and don't want to give everything AI to control all the output

2B · BRINGING THREE SPECIALISTS ALONG

Assess with an artefact, not an opinion. Give all three the same task on this dataset — clean it, forecast one language, document the assumptions

AI is a helping hand. I would try to understand, how the team currently uses AI, is it prompt driven? Do they create a plan before execution? How do they validate the output of the AI? With AI we need to understand when we need to give it full control and we need to take back the control. E.g. giving all the numbers to AI and asking for plan: All control is given to the AI whereas if you create a framework through which you can input any numbers and get a plan, this is where you have full control. Setting up best practices on AI usage is also will be my focus

Strengths and Improvement areas. Few team members may be more technical/business oriented therefore within the team I need to make sure that whenever we get a problem statement we split into multiple sub task and assign it to the team members such that each sub task is relevant for each team member but at end we should all collaborate and share each other learning. Business oriented analyst can be shown how automation works and technical analyst can be shown how business decisions are taken

2B · MEASURING SUCCESS

To measure the transformation is on track?

My KPIs would be

1. Time required to produce the forecast
2. MAPE of the forecast
3. % of automation: Let's say we have 10 components in forecast framework, how many of them are automated
4. Scenario MAPE: Not only track the accuracy of our actual forecast but different scenario which we may have discussed

In addition to KPIs, I would also look at

1. Do we have single of source truth?
2. Do we have dashboard where we daily/weekly track accuracy of our plan
3. Do we have standard process such as SQL queries by which we can fetch and check the data
4. How does system react to big deviations, what stakeholder friction is creates

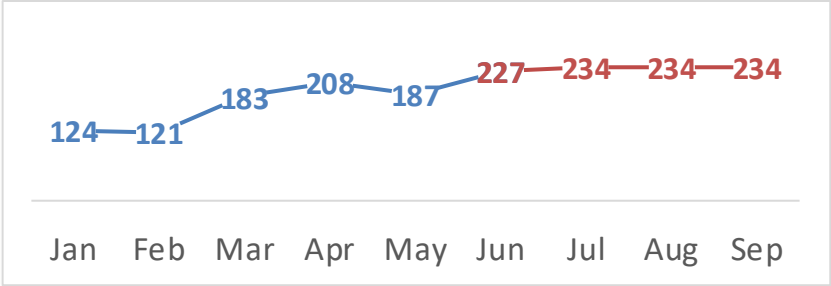
So it is not only about KPIs but also how we can streamline the process

Appendix

Forecast, capacity and the range around it

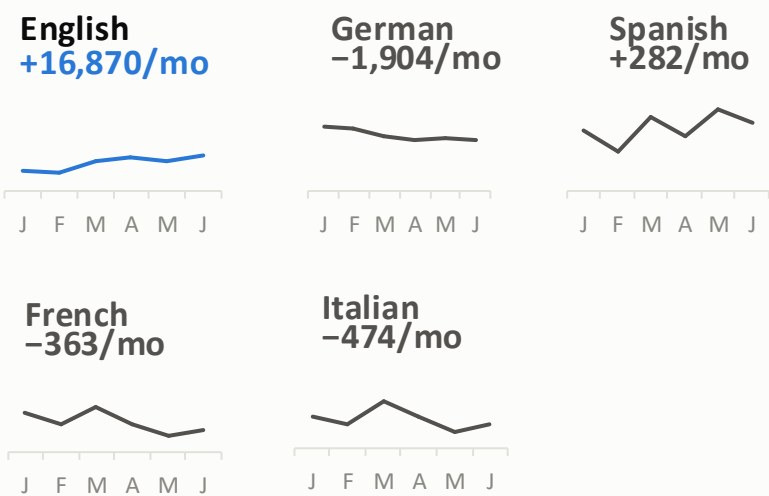
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VOLUME — SIX MONTHS OF ACTUALS, THREE OF FORECAST



BASE holds Jul–Sep flat; HIGH carries English’s own 3-month trend forward.

ONLY ENGLISH IS GROWING



2

CAPACITY BY LANGUAGE — BASE, PER

LANGUAGE	CONTACTS	PROD H	WKLD FTE	SLA FTE	PAID FTE
English	174,128	40,298	242	284	297
German	36,135	8,220	36	58	61
Spanish	32,879	7,133	31	50	53
French	26,662	6,773	26	48	50
Italian	24,066	6,402	25	45	47
Total	293,869	68,827	360	484	508

BY BPO — VENDOR SPLIT

BPO	LANGUAGES	FTE	PROD H
BPO 3	DE IT ES FR	201	28,529
BPO 1	English 50%	142	20,149
BPO 2	English 50%	142	20,149

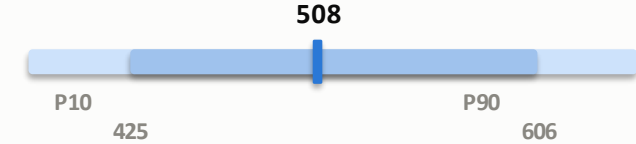
The 50/50 English split is **assumed**, English AHT is the BPO1/BPO2 blend either way, so total capacity is split-agnostic, but the per-vendor bill above isn’t.

3

SCENARIOS — PAID FTE

CASE	JUL	AUG	SEP	BASIS
BASE	508	508	508	June × 1.10, held flat
HIGH	518	551	579	English 3-mo trend ×1.10; rest ×1.10

PAID FTE — 80% CONFIDENCE BAND



Do not hire to a point estimate. Contract a ~490 FTE core and hold a cross-trained flex pool of 60–100 against the P90, released monthly on the re-forecast trigger.

Every queue, end to end — BASE, July

The full working behind Occupancy below 100% on the Erlang rows is the cost of the service level.

WHERE THE 484 FTE SITS												
LANG	CHAN	CONTACTS	AHT	MODEL	ERL	SEATS	OCC	SL	WKLD H	PROD H	WKLD FTE	SLA FTE
English	Chat	42,882	1,150	Erlang C	18.77	24	78%	86.2%	13,699	14,600	96.4	102.7
English	Email	42,882	1,150	deferred	—	—	85%	—	13,699	16,116	96.4	113.4
English	Inbound	44,182	458	Erlang C	7.69	11	70%	82.6%	5,615	8,030	39.5	56.5
English	Outbound	44,182	108	deferred	—	—	85%	—	1,319	1,552	9.3	10.9
French	Chat	6,566	900	Erlang C	2.25	5	45%	92.5%	1,641	3,042	11.5	21.4
French	Email	6,566	600	deferred	—	—	85%	—	1,094	1,287	7.7	9.1
French	Inbound	6,765	360	Erlang C	0.93	3	31%	93.3%	677	2,190	4.8	15.4
French	Outbound	6,765	115	deferred	—	—	85%	—	216	254	1.5	1.8
German	Chat	8,899	900	Erlang C	3.05	6	51%	91.3%	2,225	3,650	15.7	25.7
German	Email	8,899	700	deferred	—	—	85%	—	1,730	2,036	12.2	14.3
German	Inbound	9,168	360	Erlang C	1.26	3	42%	85.7%	917	2,190	6.5	15.4
German	Outbound	9,168	115	deferred	—	—	85%	—	293	345	2.1	2.4
Italian	Chat	5,927	900	Erlang C	2.03	4	51%	84.1%	1,482	2,433	10.4	17.1
Italian	Email	5,927	800	deferred	—	—	85%	—	1,317	1,549	9.3	10.9
Italian	Inbound	6,106	360	Erlang C	0.84	3	28%	94.8%	611	2,190	4.3	15.4
Italian	Outbound	6,106	115	deferred	—	—	85%	—	195	229	1.4	1.6
Spanish	Chat	8,097	900	Erlang C	2.77	5	55%	84.2%	2,024	3,042	14.2	21.4
Spanish	Email	8,097	600	deferred	—	—	85%	—	1,350	1,588	9.5	11.2
Spanish	Inbound	8,342	360	Erlang C	1.14	3	38%	88.7%	834	2,190	5.9	15.4
Spanish	Outbound	8,342	115	deferred	—	—	85%	—	266	314	1.9	2.2
Total		293,869							51,203	68,827	360.2	484.2

Chat is the single largest line at 39%, which is why concurrency is the biggest available lever.

Outbound is 4% despite carrying a quarter of all contacts — its AHT is 107s against email's 1,150s.

Email at 159 FTE is modelled as deferred work at 85% planned occupancy — an explicit assumption, not from the workbook.

The deferred rows carry no Erlang figures by design: a 120-minute target is a backlog commitment, not a queue discipline, so seats are derived from hours and planned occupancy instead.

Data-quality log — all 7 checks

Condensed to five rows on slide 3. This is the full log, including the checks that passed.

#	STATUS	CHECK	FINDING	DECISION
1	FIX	CSAT reported against zero tickets	3 reason-months carry a CSAT score with 0 tickets	null the CSAT; keep the row
2	FIX	reason label missing	blank label carrying 3,729 tickets (0.63% of EN volume)	relabel 'UNMAPPED' and keep the volume
3	FIX	AHT outliers vs reason median	1 reason-month outside 0.5x–2.0x its median	replace with the reason's own median
4	WARN	channel pair Inbound == Outbound	identical in 29/30 language-months	synthetic mirror — kept conservatively, flagged
5	WARN	channel pair Email == Chat	identical in 23/30 language-months	synthetic mirror — kept conservatively, flagged
6	FIX	SLA attainment series within [0,1]	February reports 110% attainment	exclude the impossible month
7	WARN	SLA attainment trend	95%→92% while volume grows	use as the business case for the transformation

Assumptions

EVERY ASSUMPTION, AND WHERE IT COMES FROM

ASSUMPTION	VALUE	NOTE
Hours per FTE per month	173.33	=40 × 52 ÷ 12
Total shrinkage	18%	divisor 0.82
Operating hours per month	730	24/7 → 8,760 ÷ 12
Phone SLA	80% ≤ 20 s	Erlang C
Chat SLA	80% ≤ 60 s	Erlang C, ÷ concurrency
Email SLA	80% ≤ 120 min	deferred model
Chat concurrency	1.2	open question — worth 75 FTE
Deferred occupancy	85%	modelling assumption — not in the data
Seasonal uplift	+10%	applied to the June baseline
Annual attrition	36%	compounds to 3.65%/month
Learning curve	120/110/105%	AHT multiplier, months 1–3
Training	4 weeks	non-producing, paid
Routing	BPO 1&2 EN; BPO 3 DE/IT/FR/ES	9 of 15 AHT rows unroutable
English BPO split	50 / 50	AHT blended — assumption
English AHT source	channel-level (709.8s)	vs reason-level 908.8s — worth 78 FTE